Worksheet 5: Multiqubits and Quantum Circuits

Solutions

Quantum Computation and Error Correction
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Discussed on October 20, 2026.

Conventions are those of the question sheet: qubits are numbered from 0, $|ab\rangle$ means qubit 0 is in state a and qubit 1 in state b, and matrices are written in the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. In a tensor product $A \otimes B$ the first factor acts on qubit 0. $P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$. A circuit "first G_1, then G_2" is the matrix $G_2 G_1$, which is the single most common source of sign-of-time mistakes on this sheet, so keep it in mind. Please let me know if you find an error.

Part A — Two qubits: states and gates as matrices

1. Tensor products in practice

(a) The basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ means $|01\rangle$ is the second basis vector:

$$|01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

For $|+\rangle \otimes |-\rangle$, the Kronecker product takes each component of the first vector and multiplies it into a copy of the second:

$$|+\rangle \otimes |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}.$$

So the amplitudes are $c_{00} = c_{10} = \frac{1}{2}$, $c_{01} = c_{11} = -\frac{1}{2}$.

(b) Following the hint, put the four amplitudes into the matrix $C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix}$. If $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ with $|\alpha\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $|\beta\rangle = \beta_0|0\rangle + \beta_1|1\rangle$, then multiplying out gives $c_{ab} = \alpha_a \beta_b$, which says

$$C = \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \begin{pmatrix} \beta_0 & \beta_1 \end{pmatrix} = \alpha \beta^T.$$

A matrix of this form (an outer product) has rank one, so its determinant vanishes: $c_{00}c_{11} - c_{01}c_{10} = 0$. That is one direction.

For the other direction, suppose $c_{00}c_{11} = c_{01}c_{10}$, i.e. $\det C = 0$. The state is normalised so $C \neq 0$, and a non-zero 2×2 matrix with zero determinant has rank one. Rank one means the two columns are proportional: there is a vector $u = (u_0, u_1)^T$ and numbers v_0, v_1 such that the first column is $v_0 u$ and the second is $v_1 u$, which is to say $c_{ab} = u_a v_b$. But then $|\psi\rangle = \sum_{ab} u_a v_b |ab\rangle = \big(\sum_a u_a |a\rangle\big) \otimes \big(\sum_b v_b |b\rangle\big)$, a product state. (Normalise the two factors separately if you want them to be unit vectors; the overall normalisation of $|\psi\rangle$ guarantees this can be done.)

So the whole business of "is this two-qubit state entangled?" reduces to one determinant.

(c) Apply the test to each state.

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: $C = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, $\det C = \frac{1}{2} \neq 0$. Entangled; this is the first Bell state, and it is not a product state however you try to write it.

$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$: all four amplitudes are $\frac{1}{2}$, so $c_{00}c_{11} = c_{01}c_{10} = \frac{1}{4}$. Product state, and in fact you can see the factorisation directly: it is $\frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) = |+\rangle \otimes |+\rangle$.

$\frac{1}{2}(|00\rangle+|01\rangle-|10\rangle-|11\rangle)$: $c_{00}c_{11} = \frac{1}{2}\cdot(-\frac{1}{2}) = -\frac{1}{4}$ and $c_{01}c_{10} = \frac{1}{2}\cdot(-\frac{1}{2}) = -\frac{1}{4}$. Equal, so product. The sign goes with the value of qubit 0 (it is $-$ exactly when $a = 1$), so the first factor is $|-\rangle$ and the second is $|+\rangle$: the state is $|-\rangle\otimes|+\rangle$. Check: $\frac{1}{2}(|0\rangle-|1\rangle)(|0\rangle+|1\rangle) = \frac{1}{2}(|00\rangle+|01\rangle-|10\rangle-|11\rangle)$. Good.

(d) "H on qubit 0, then X on qubit 1" is the matrix $(\mathbb{I}\otimes X)(H\otimes\mathbb{I})$, and by the rule $(A\otimes B)(C\otimes D) = AC \otimes BD$ this equals $H \otimes X$. The other order is $(H \otimes \mathbb{I})(\mathbb{I} \otimes X) = H \otimes X$ too. Same matrix. The point is that $\mathbb{I}$ commutes with everything, so when the two gates sit on different wires there is nothing for them to disagree about, and that is why a circuit diagram can put them in the same column without saying which goes first.

2. The two workhorses: `cx` and `cz`

(a) $\texttt{cx}(0,1)$ leaves $|00\rangle$ and $|01\rangle$ alone and swaps $|10\rangle \leftrightarrow |11\rangle$. The columns of the matrix are the images of the basis vectors, so

$$\texttt{cx}(0,1) = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}, \qquad \texttt{cx}(1,0) = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\end{pmatrix},$$

where for $\texttt{cx}(1,0)$ it is $|01\rangle \leftrightarrow |11\rangle$ that get exchanged (qubit 1 is 1 in both, and qubit 0 gets flipped). They are permutation matrices because the gate is defined by what it does to classical bit strings: each basis state goes to exactly one basis state, with no superposition and no phase, and the map is reversible. The same is true of any gate that comes from classical reversible logic, like the Toffoli from Worksheet 1.

(b) Act with $P_0 \otimes \mathbb{I} + P_1 \otimes X$ on a basis state $|ab\rangle$. If $a = 0$, the second term kills it (since $P_1|0\rangle = 0$) and the first term leaves it alone: $|0b\rangle \mapsto |0b\rangle$. If $a = 1$, the first term kills it and the second applies X to qubit 1: $|1b\rangle \mapsto |1,\, b \oplus 1\rangle$. That is exactly the `cx` rule, so the two operators agree on a basis and are equal. In words the formula reads: "project the control onto 0 and do nothing, plus project the control onto 1 and flip the target". For the other direction the projectors sit on qubit 1 and the X on qubit 0:

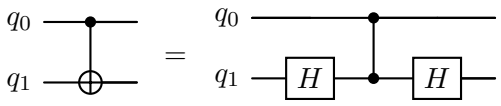
$$\texttt{cx}(1,0) = \mathbb{I} \otimes P_0 + X \otimes P_1.$$

(c) $\texttt{cz}(0,1) = P_0 \otimes \mathbb{I} + P_1 \otimes Z$ by the same reading: nothing if the control is 0, a Z on the target if it is 1. As a matrix, $P_0 \otimes \mathbb{I} = \mathrm{diag}(1,1,0,0)$ and $P_1 \otimes Z = \mathrm{diag}(0,0,1,-1)$, which add to $\mathrm{diag}(1,1,1,-1)$. Doing the same with the roles exchanged, $\texttt{cz}(1,0) = \mathbb{I} \otimes P_0 + Z \otimes P_1 = \mathrm{diag}(1,0,1,0) + \mathrm{diag}(0,1,0,-1) = \mathrm{diag}(1,1,1,-1)$ as well. So $\texttt{cz}(0,1) = \texttt{cz}(1,0)$. This was bound to happen, because `cz` multiplies $|ab\rangle$ by $(-1)^{ab}$, and ab is symmetric in a and b: there is simply no asymmetry in the definition for a control/target distinction to latch onto. Not so for `cx`: the matrices in (a) are visibly different, and already on $|01\rangle$ one does nothing and the other gives $|11\rangle$.

(d) Conjugating a tensor product termwise,

$$\begin{aligned}(\mathbb{I}\otimes H)\,\texttt{cz}(0,1)\,(\mathbb{I}\otimes H) &= (\mathbb{I}\otimes H)(P_0\otimes\mathbb{I} + P_1\otimes Z)(\mathbb{I}\otimes H)\\ &= P_0 \otimes H\mathbb{I}H + P_1 \otimes HZH = P_0\otimes\mathbb{I} + P_1\otimes X = \texttt{cx}(0,1),\end{aligned}$$

using $H^2 = \mathbb{I}$ and $HZH = X$. As circuits,



One way to say it: `cz` is a `cx` that flips the target in the X basis (Z swaps $|+\rangle$ and $|-\rangle$), and the two Hadamards translate between the X basis and the Z basis on that wire.

(e) The first one is the standard calculation,

$$\texttt{cx}(0,1)\,|+\rangle|0\rangle = \tfrac{1}{\sqrt{2}}\,\texttt{cx}(0,1)\big(|00\rangle + |10\rangle\big) = \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big),$$

a Bell state; the target has become entangled with the control. For the second, use that $|-\rangle$ is an eigenvector of X: $X|-\rangle = -|-\rangle$. So on each branch of the control,

$$\texttt{cx}(0,1)\,|a\rangle|-\rangle = |a\rangle \otimes X^a|-\rangle = (-1)^a|a\rangle|-\rangle,$$

and therefore

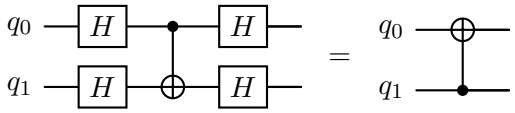
$$\texttt{cx}(0,1)\,|+\rangle|-\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle - |1\rangle\big)|-\rangle = |-\rangle|-\rangle.$$

The target really is untouched, and the control has gone from $|+\rangle$ to $|-\rangle$. There is no contradiction with "flips qubit 1 if qubit 0 is 1": that sentence describes what happens to basis states, and on each basis state of the control it is true. The catch is that flipping $|-\rangle$ does not produce a new state but the same state with a minus sign, and when the control is in a superposition that minus sign is attached to the $|1\rangle$ branch only. A sign on one branch is a relative phase, and a relative phase on the control is a change of the control. The gate did precisely what it says; it is our habit of thinking of "the target changes" as something visible on the target that misleads. This phenomenon is called phase kick-back, and it is used all over the place (Deutsch–Jozsa, phase estimation, Shor).

Part B — Circuit identities

3. Hadamards turn a `cx` around

We want to show



without touching a 4×4 matrix.

(a) Multiply out, with $H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$:

$$HX = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix},$$

and then

$$HXH = \frac{1}{2}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2 & 0\\ 0 & -2\end{pmatrix} = Z.$$

Then $HZH = H(HXH)H = H^2XH^2 = X$, no second calculation needed. (If you prefer pictures: H exchanges the x and z axes of the Bloch sphere, and X, Z are the half turns about those axes.)

(b) Substitute $P_0 = \frac{1}{2}(\mathbb{I} + Z)$, $P_1 = \frac{1}{2}(\mathbb{I} - Z)$ into Q2(b):

$$\texttt{cx}(0,1) = \tfrac{1}{2}(\mathbb{I} + Z) \otimes \mathbb{I} + \tfrac{1}{2}(\mathbb{I} - Z) \otimes X = \tfrac{1}{2}\big(\mathbb{I} \otimes \mathbb{I} + Z \otimes \mathbb{I} + \mathbb{I} \otimes X - Z \otimes X\big).$$

(c) Conjugation by $H \otimes H$ acts on each factor separately, $(H \otimes H)(A \otimes B)(H \otimes H) = HAH \otimes HBH$, and by (a) it just swaps the letters $X \leftrightarrow Z$ while leaving $\mathbb{I}$ alone. Term by term:

$$\mathbb{I} \otimes \mathbb{I} \mapsto \mathbb{I} \otimes \mathbb{I}, \qquad Z \otimes \mathbb{I} \mapsto X \otimes \mathbb{I}, \qquad \mathbb{I} \otimes X \mapsto \mathbb{I} \otimes Z, \qquad Z \otimes X \mapsto X \otimes Z.$$

So

$$\begin{aligned}(H \otimes H)\,\texttt{cx}(0,1)\,(H \otimes H) &= \tfrac{1}{2}\big(\mathbb{I} \otimes \mathbb{I} + X \otimes \mathbb{I} + \mathbb{I} \otimes Z - X \otimes Z\big) \\ &= \mathbb{I} \otimes \tfrac{1}{2}(\mathbb{I} + Z) + X \otimes \tfrac{1}{2}(\mathbb{I} - Z) = \mathbb{I} \otimes P_0 + X \otimes P_1,\end{aligned}$$

and the last expression is $\texttt{cx}(1,0)$ by Q2(b). Done, and notice how the regrouping works: everything with $\mathbb{I}$ on qubit 0 collects into $\mathbb{I} \otimes P_0$, everything with X on qubit 0 into $X \otimes P_1$.

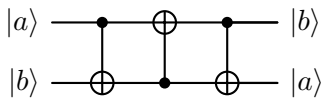
(d) What (c) says is that in the basis $\{|{+}{+}\rangle, |{+}{-}\rangle, |{-}{+}\rangle, |{-}{-}\rangle\}$, i.e. the basis $(H \otimes H)|ab\rangle$, the gate $\texttt{cx}(0,1)$ looks like $\texttt{cx}(1,0)$. Write $|x,y\rangle_X$ for $|\pm\rangle \otimes |\pm\rangle$ with $x, y \in \{0,1\}$ coding $+$ as 0 and $-$ as 1. Then

$$\texttt{cx}(0,1)\,|x,y\rangle_X = |x \oplus y,\ y\rangle_X :$$

qubit 1 is unchanged and its value is added to qubit 0. So the answer is the X basis, the one made of $|\pm\rangle$. This is Q2(e) again in a more systematic form: $|+\rangle|-\rangle$ is $|0,1\rangle_X$, which goes to $|1,1\rangle_X = |-\rangle|-\rangle$. In the Z basis the information flows from qubit 0 to qubit 1; in the X basis it flows the other way. A `cx` has a control and a target only once you have chosen a basis.

4. Three `cx` gates make a swap

(a) Follow $|ab\rangle$ through the circuit



one gate at a time:

$$|a,b\rangle \xrightarrow{\texttt{cx}(0,1)} |a,\ a \oplus b\rangle \xrightarrow{\texttt{cx}(1,0)} |a \oplus (a \oplus b),\ a \oplus b\rangle = |b,\ a \oplus b\rangle \xrightarrow{\texttt{cx}(0,1)} |b,\ b \oplus (a \oplus b)\rangle = |b,a\rangle.$$

The only algebra is $a \oplus a = 0$, used twice. If you know the classical trick for swapping two variables with three XORs and no temporary variable, this is exactly that.

(b) Both the circuit and `swap` are linear maps, and a linear map is completely determined by what it does to a basis: for a general state $|\psi\rangle = \sum_{ab} c_{ab}|ab\rangle$ we get $U|\psi\rangle = \sum_{ab} c_{ab}\,U|ab\rangle$, so if U and `swap` agree on the four $|ab\rangle$ they agree on every $|\psi\rangle$. Entangled states are not special here; they are just linear combinations of basis states like any other. (The same remark justifies "it is enough to check basis states" everywhere on this sheet, and it is worth being clear about exactly what is being used: linearity, nothing more.)

(c) There are only four circuits with two `cx` gates: the two directions, each followed by either direction. Two of them use the same gate twice and are the identity, because $\texttt{cx}^2 = \mathbb{I}$ (flipping the target twice under the same condition undoes itself). The identity is not a swap, since $\texttt{swap}|01\rangle = |10\rangle \neq |01\rangle$. The other two are



Left circuit, $\texttt{cx}(0,1)$ then $\texttt{cx}(1,0)$: $|01\rangle \mapsto |01\rangle \mapsto |11\rangle$, but a swap should give $|10\rangle$. Right circuit, $\texttt{cx}(1,0)$ then $\texttt{cx}(0,1)$: $|01\rangle \mapsto |11\rangle \mapsto |10\rangle$, which looks promising, but $|10\rangle \mapsto |10\rangle \mapsto |11\rangle$ instead of $|01\rangle$. So none of the four is a swap, and three `cx` gates is the minimum.

5. Pushing errors through a `cx`

Throughout, `cx` means $\texttt{cx}(0,1)$ and I use the projector form $\texttt{cx} = P_0 \otimes \mathbb{I} + P_1 \otimes X$ from Q2(b). The useful one-qubit facts are $XP_0X = P_1$, $XP_1X = P_0$ (equivalently $XP_0 = P_1X$, $XP_1 = P_0X$), $ZP_0 = P_0$, $ZP_1 = -P_1$ and $XZX = -Z$.

(a) Push $X \otimes \mathbb{I}$ through the first `cx` from the left:

$$(X \otimes \mathbb{I})\,\texttt{cx} = XP_0 \otimes \mathbb{I} + XP_1 \otimes X = P_1X \otimes \mathbb{I} + P_0X \otimes X = \big(P_1 \otimes \mathbb{I} + P_0 \otimes X\big)(X \otimes \mathbb{I}).$$

Now multiply by `cx` on the left. Cross terms contain $P_0P_1 = 0$ and drop out; the surviving ones use $P_a^2 = P_a$ and $X^2 = \mathbb{I}$:

$$\texttt{cx}\,\big(P_1 \otimes \mathbb{I} + P_0 \otimes X\big) = P_0 \otimes X + P_1 \otimes X = \mathbb{I} \otimes X.$$

Hence $\texttt{cx}\,(X \otimes \mathbb{I})\,\texttt{cx} = (\mathbb{I} \otimes X)(X \otimes \mathbb{I}) = X \otimes X$. The bit flip on the control has been copied onto the target. Intuitively this is obvious: if the control was wrongly flipped, the target gets flipped exactly when it should not have been, which is the same as flipping it on top of the correct result.

(b) Here Z sits on the target and commutes with the projectors on the control, so sandwiching is easy:

$$\texttt{cx}\,(\mathbb{I} \otimes Z)\,\texttt{cx} = P_0 \otimes Z + P_1 \otimes XZX = P_0 \otimes Z - P_1 \otimes Z = (P_0 - P_1) \otimes Z = Z \otimes Z.$$

A phase flip on the target has been copied onto the control. This is (a) in the X basis, which is what Q3 would lead you to expect: Z is the bit flip of the X basis, and in the X basis qubit 1 is the control.

(c) X on the target:

$$\texttt{cx}\,(\mathbb{I} \otimes X)\,\texttt{cx} = P_0 \otimes X + P_1 \otimes XXX = P_0 \otimes X + P_1 \otimes X = \mathbb{I} \otimes X.$$

Z on the control: Z commutes with P_0 and P_1, so

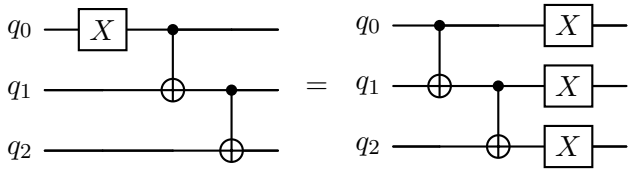
$$\texttt{cx}\,(Z \otimes \mathbb{I})\,\texttt{cx} = P_0Z \otimes \mathbb{I} + P_1Z \otimes X^2 = P_0 \otimes \mathbb{I} - P_1 \otimes \mathbb{I} = Z \otimes \mathbb{I}.$$

In both cases the error goes straight through. Summary of the four cases, which is worth memorising:

error before `cx`	error after `cx`	
X on control	X on control and target	copied
Z on target	Z on control and target	copied
X on target	X on target	unchanged
Z on control	Z on control	unchanged

X errors travel from control to target; Z errors travel from target to control.

(d) Start with $X \otimes \mathbb{I} \otimes \mathbb{I}$. The first gate, $\texttt{cx}(0,1)$, has the error on its control, so by (a) it becomes $X \otimes X \otimes \mathbb{I}$. The second gate, $\texttt{cx}(1,2)$, does not touch qubit 0, and the X on qubit 1 is on its control, so by (a) again it gets copied to qubit 2: $X \otimes X \otimes X$. One bit flip has turned into three. In pictures, the left circuit equals the right one:



The reason we care: if the circuit is part of an error-correcting code, a single fault on one qubit that spreads to three qubits may be beyond what the code can fix. Fault-tolerant circuit design is largely about arranging gates so that this does not happen, and the bookkeeping is exactly the four rules in the table. The same bookkeeping, run in the other direction, is how a syndrome measurement tells you where an error was.

Part C — Entanglement from circuits

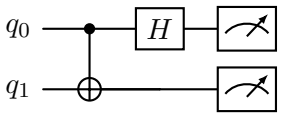
6. The Bell circuit

(a) The circuit is H on qubit 0 followed by $\texttt{cx}(0,1)$. The Hadamard turns the first qubit into $|\pm\rangle$ and the `cx` then copies the Z-basis value onto the second qubit:

$$|00\rangle \mapsto |+\rangle|0\rangle \mapsto \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad |01\rangle \mapsto |+\rangle|1\rangle \mapsto \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$
$$|10\rangle \mapsto |-\rangle|0\rangle \mapsto \tfrac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \qquad |11\rangle \mapsto |-\rangle|1\rangle \mapsto \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

Each is normalised, and any two are orthogonal: either they are built on disjoint pairs of basis states (first and second, say) or on the same pair with opposite relative signs (first and third), and in the latter case the inner product is $\frac{1}{2}(1-1) = 0$. You could also just say: a unitary applied to an orthonormal basis gives an orthonormal basis. For entanglement, apply Q1(b): the first and third have $c_{00}c_{11} = \pm\frac{1}{2}$ but $c_{01}c_{10} = 0$; the second and fourth have $c_{01}c_{10} = \pm\frac{1}{2}$ but $c_{00}c_{11} = 0$. The determinant never vanishes, so none of the four is a product state.

(b) Reverse the order of the gates and invert each one. Both H and `cx` are their own inverses, so the circuit is simply $\texttt{cx}(0,1)$ first, then H on qubit 0:



Run on $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ it gives $|+\rangle|0\rangle$ and then $|00\rangle$, and so on for the others. Measuring both qubits in the Z basis afterwards tells you which Bell state you had: 00, 01, 10, 11 in the order of (a). That is a Bell measurement, and it is how teleportation and dense coding are actually implemented.

(c) Expand in the Z basis and watch the cross terms cancel:

$$|{+}{+}\rangle + |{-}{-}\rangle = \tfrac{1}{2}\big[(|0\rangle+|1\rangle)(|0\rangle+|1\rangle) + (|0\rangle-|1\rangle)(|0\rangle-|1\rangle)\big] = \tfrac{1}{2}\big[2|00\rangle + 2|11\rangle\big] = |00\rangle + |11\rangle.$$

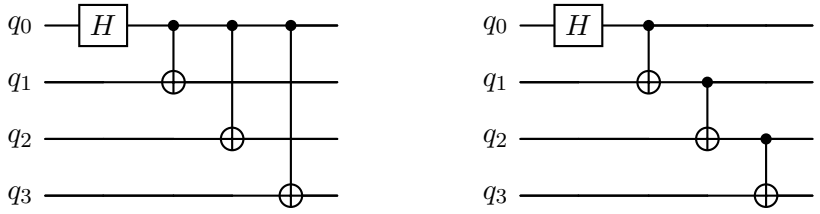
Dividing by $\sqrt{2}$ gives the claim. From the Z-basis form, measuring both qubits in the Z basis gives 00 or 11 with probability $\frac{1}{2}$ each, never 01 or 10. From the X-basis form, measuring both in the X basis gives $++$ or $--$ with probability $\frac{1}{2}$ each, never $+-$ or $-+$. The outcomes are individually random and jointly perfectly correlated, in both bases at once.

(d) Suppose $|\alpha\rangle \otimes |\beta\rangle$ gives agreeing results with certainty in the Z basis. The probability of the outcome 01 is $|\alpha_0|^2|\beta_1|^2$ and of 10 is $|\alpha_1|^2|\beta_0|^2$, and both must be zero. If $\alpha_0 \neq 0$ the first forces

$\beta_1 = 0$, so $|\beta\rangle = |0\rangle$ up to phase, and then the second forces $\alpha_1 = 0$, so $|\alpha\rangle = |0\rangle$ as well. If instead $\alpha_0 = 0$ then $|\alpha\rangle = |1\rangle$ and the second condition forces $\beta_0 = 0$, $|\beta\rangle = |1\rangle$. Either way the state is $|00\rangle$ or $|11\rangle$ up to a phase. Now measure these in the X basis: $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$, and likewise for $|1\rangle$ with a minus sign, so each qubit gives $+$ or $-$ with probability $\frac{1}{2}$, independently of the other. They agree only half the time. So there is no product state that does what the Bell state does in (c). This is a baby version of a Bell inequality: perfect correlations in two incompatible bases are something only entanglement can deliver.

7. GHZ states: fan-out versus chain

The two circuits are



Fan-out on the left, chain on the right.

(a) After the Hadamard the state is $\frac{1}{\sqrt{2}}(|0000\rangle + |1000\rangle)$. The $|0000\rangle$ branch is untouched by every `cx` in either circuit, since every control is 0. On the $|1000\rangle$ branch the fan-out copies the 1 from qubit 0 onto qubits 1, 2, 3 in turn: $|1000\rangle \to |1100\rangle \to |1110\rangle \to |1111\rangle$. The chain passes it down the line instead, $\texttt{cx}(0,1)$ gives $|1100\rangle$, then $\texttt{cx}(1,2)$, with qubit 1 now 1, gives $|1110\rangle$, then $\texttt{cx}(2,3)$ gives $|1111\rangle$. Both end at

$$\tfrac{1}{\sqrt{2}}\big(|0000\rangle + |1111\rangle\big).$$

(b) Take the input $|0100\rangle$. After H: $\frac{1}{\sqrt{2}}(|0100\rangle + |1100\rangle)$. Fan-out: the $|0100\rangle$ branch is untouched; the other becomes $|1000\rangle \to |1010\rangle \to |1011\rangle$. Output $\frac{1}{\sqrt{2}}(|0100\rangle + |1011\rangle)$. Chain: in the $|0100\rangle$ branch the first `cx` does nothing, but $\texttt{cx}(1,2)$ now fires because qubit 1 is 1, giving $|0110\rangle$, and then $\texttt{cx}(2,3)$ gives $|0111\rangle$; the other branch becomes $|1000\rangle$ and stays there. Output $\frac{1}{\sqrt{2}}(|0111\rangle + |1000\rangle)$. The two outputs are different, so the two unitaries are different, even though they agreed perfectly on $|0000\rangle$.

Any input with a 1 on qubit 1 or qubit 2 would have done. The two circuits agree on a basis input exactly when qubits 1 and 2 start at 0 ($|0000\rangle$, $|0001\rangle$, $|1000\rangle$, $|1001\rangle$), because then every control in the chain carries the same value as the control in the fan-out, and on the other twelve inputs they disagree. (Qubit 3 is only ever a target, so its initial value does not matter.) The moral stands: checking one input tells you nothing about equality of circuits; checking a full basis does, by Q4(b).

(c) Write the GHZ state as $\frac{1}{\sqrt{2}}(|0\rangle|000\rangle + |1\rangle|111\rangle)$. Measuring qubit 0 in the Z basis gives 0 or 1 with probability $\frac{1}{2}$ each, and the other three qubits are left in $|000\rangle$ or $|111\rangle$ respectively. Both are product states: no entanglement is left. Reading one qubit of a GHZ state in the Z basis tells you the values of all the others and destroys the superposition completely.

(d) Substitute $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$ for qubit 0 and collect terms:

$$\tfrac{1}{\sqrt{2}}\big(|0\rangle|000\rangle + |1\rangle|111\rangle\big) = \tfrac{1}{2}\Big[|+\rangle\big(|000\rangle + |111\rangle\big) + |-\rangle\big(|000\rangle - |111\rangle\big)\Big].$$

The two three-qubit states in brackets each have norm $\sqrt{2}$, so each outcome has probability $\frac{1}{4} \cdot 2 = \frac{1}{2}$, and after the measurement the remaining qubits are in

$$\tfrac{1}{\sqrt{2}}\big(|000\rangle \pm |111\rangle\big),$$

with the + for outcome + and the − for outcome −. These are three-qubit GHZ states, and they are entangled: splitting off qubit 1, say, the coefficient matrix in the sense of Q1(b) (now 2×4, rows indexed by qubit 1, columns by qubits $2, 3$) has non-zero entries only at $(0, 00)$ and $(1, 11)$, which is rank two, so no factorisation exists. The outcome − can be turned into + by a Z on any one of the remaining qubits, so up to a known local correction the three qubits are still in a GHZ state.

The contrast with (c) is the whole point. A Z measurement on one qubit asks "which branch, $|000\rangle$ or $|111\rangle$?" and the answer collapses everyone else. An X measurement asks about $|0\rangle \pm |1\rangle$, which is the same question for both branches, so it learns nothing about which branch and only imprints a sign on the relative phase. The entanglement among the other qubits survives. This is why GHZ-type states, and cluster states built from them, are useful: you can measure qubits away one at a time in the X basis without wrecking the rest.

8. On a computer (optional)

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Operator

# Q3: Hadamards turn a cx around
a = QuantumCircuit(2); a.h([0, 1]); a.cx(0, 1); a.h([0, 1])
b = QuantumCircuit(2); b.cx(1, 0)
print(Operator(a).equiv(Operator(b)))        # True

# Q4(a): three cx make a swap
c = QuantumCircuit(2); c.cx(0, 1); c.cx(1, 0); c.cx(0, 1)
d = QuantumCircuit(2); d.swap(0, 1)
print(Operator(c).equiv(Operator(d)))        # True

# Q7(b): fan-out versus chain
fan = QuantumCircuit(4); fan.h(0); fan.cx(0, 1); fan.cx(0, 2); fan.cx(0, 3)
chn = QuantumCircuit(4); chn.h(0); chn.cx(0, 1); chn.cx(1, 2); chn.cx(2, 3)
print(Operator(fan).equiv(Operator(chn)))    # False
```

The three prints give `True`, `True`, `False`, as they should. If you also print output states with `Statevector`, remember that Qiskit writes qubit 0 as the rightmost character of the bit string, so our $|0100\rangle$ appears as `0010` there.